AYOUB TOUJANI

Computer Science Engineer: Data & Business Intelligence

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PROFESSIONAL SUMMARY

Data and Business Intelligence Engineer with experience in data management, analytics, automation, and AI solutions within the banking and telecommunications sectors. Experienced in designing end-to-end data platforms, automating business processes, and developing dashboards that support operational and strategic decision-making. Specialized in **Data Engineering, Business Intelligence, Predictive Analytics, and Artificial Intelligence**.

PROFESSIONAL EXPERIENCE

SIBTEL - Interbank Clearing Company of Tunisia Dec. 2025 – June 2026
Automation & Data Engineer *Tunis, Tunisia*

- Designed and deployed, the first decision-support platform at SIBTEL, the sole interbank clearing operator in Tunisia (+**50 million transactions/year**, 28 member institutions).
- Built a single geographic reference for **2,200+ bank branches** through **3 n8n workflows** merging 3 different sources (internal Oracle system, Central Bank API, national postal reference data), replacing a fully manual process and giving SIBTEL its first-ever geographic map of the network.
- Automated the daily collection, cleaning, and enrichment of data (duplicate/conflict resolution, geocoding), with zero manual work needed since going live.
- Orchestrated a daily **ELT** pipeline with **Apache Airflow** feeding a **star-schema** data warehouse, protected by **40 automated quality tests** that block any bad data before it reaches users.
- Built **6 self-service Metabase dashboards** for all bank members : *National Daily Overview*, *Bank Comparison*, *Agency Network* (geographic scoring of 2,200+ branches), *Payment Analytics*, *Bank & Agency 360°*, and *District & Regional Performance* cutting analysis time from several days to under 15 minutes.
- Deployed a full **monitoring** stack (Prometheus, Grafana, 3 exporters, 8 alert rules) covering the pipeline, warehouse, database, and server, to keep the whole platform reliable end to end.
- Turned business needs from the Data Analytics, BI & Business Development unit into technical specifications and architecture choices, validated during weekly reviews with management.

Tunisie Telecom June 2025 – Sept. 2025
Business Intelligence Engineer *Tunis, Tunisia*

- Designed and deployed, a business intelligence solution for the Regional Operation Center, turning a fully reactive incident process spread across **52 monthly CSV files** (4 years of history) into one centralized, proactive view.
- Modeled (**Kimball** method, star schema: 1 fact table, 5 dimensions) and built a **PostgreSQL data warehouse**, fed by an **ETL pipeline in Talend Open Studio** that included SQL-based cleanup of the service and problem-type fields.
- Built **3 Power BI dashboards**: *Incident Management* (executive view and 4-year trends), *Problem / Root Cause Analysis* (spotting recurring causes), and *Site & Service Performance* (resolution time by site and priority) - the company's first-ever consolidated view of incident performance.
- Trained a **machine learning** model to predict how long an incident will take to resolve, based on its type, service, priority, and site, and exposed it through a **Flask API** to help plan resources ahead of time.
- Built an **Angular** web app bringing the Power BI dashboards and the prediction model together in one portal.

EDUCATION

National Engineering Degree in Computer Science 2023 – 2026
Private Higher School of Engineering and Technologies (ESPRIT) - Tunisia

Bachelor's Degree in Computer Science 2020 – 2023
ISAMM - University of Manouba, Tunisia

KEY PROJECTS

EDvance: Admission and Employability BI & AI Solution Jan. – May 2025 GitHub
Designed and implemented a complete **business intelligence & AI** solution to optimize the management of the admission process for students and track their performance to ensure to them better employability once they graduated.

SmartPredict : AI Smartphone Recommendation System Sep. – Dec. 2024 GitHub
Developed a machine-learning solution to help users select smartphones based on their needs and budget.

SKILLS & LANGUAGES

Data Engineering	Apache Airflow (DAGs, scheduling), dbt (tests, incremental models), n8n, Talend Open Studio, SSIS
Data Warehousing	PostgreSQL, TimescaleDB, MySQL, Oracle, SQL Server, Star schema, Kimball
BI & Visualization	Power BI (DAX, Power Query), Metabase, Grafana + Prometheus, KPI reporting, data quality
Analytics & AI	Python (Pandas, scikit-learn), Machine Learning, Flask API, Angular, supervised/unsupervised learning
DevOps & Tools	Docker Compose, Git, Microsoft Power Apps, SAP Fiori, Tesseract OCR, Advanced Excel
Languages	**French:** C1 | **English:** C1 | **Arabic:** Native